

Tianshuo WEI

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Education Background

- 2025.09–Now **Phd Data Science**, *City University of Hong Kong*, Supervisor: **Kaidi XU**
- 2024.09– **MSc Data Science**, *City University of Hong Kong*, Supervisor: **Xiangyu Zhao**
- 2025.10
 - Course: Deep Learning, Statistical Machine Learning, Dynamic Programming and Reinforcement Learning, Exploratory Data Analysis and Visualization
 - Average GPA: 3.9/4.3. Distinction.
- 2016.09– **BSc in Mathematics and Applied Mathematics**, *University of Nottingham*
- 2020.06
 - Degree: Upper Second Class Honours (2:1)

Research Interest

Memory of LLM, Embodied AI, Large Language Models, AI for Science

Research Projects

- 2025.10–Now **Personal Research Assistant**, *Prof.Kaidi XU Team*
- Research projects
 - Recent project aims to explore the emerging security challenges brought by **Embodied AI** driven by **Large Language Models (LLMs)**, namely the physical risks that attacks in the digital world may lead to in the real world, such as AI perception (computer vision, sensors), AI decision-making (behavioral decision-making, human-computer interaction), and so on.
- 2025.06– **Personal Research Assistant**, *Applied Computational Intelligence*
- 2025.11
 - Lead the development of **LLM agents** for scientific discovery, focusing on solving complex problems in fields such as **Inequality** and **Materials Design**.
 - Using LLM to analyze and interpret multimodal temporal data (such as EEG, eye tracking) aims to extract deep human intentions and system states from complex physical world interactions.
- 2024.09– **Personal Research Assistant**, *Applied Machine Learning Lab*
- 2025.10

Mainly researching on **Recommender System** and **LLM**. Successfully completed master's degree and obtained first published article under the guidance of *Prof.Zhao* and *Prof.Wang*.
- 2023.10– **Research Assistant**, *Wangxuan Institute of Computer Technology, Peking University*
- 2024.08

Develop a high-performance and intelligent Chinese text correction system based on Large Language Model (LLM)

 - Data Engineering and Mining: Solve the problem of data scarcity
 - Model pre training and architecture: Continuous pre training was implemented on the Qwen-7B model.

Publication

- Publication
- Tianyi Xu, **Tianshuo Wei**, Ye Wei.(2026). *Sustainable Materials Design with Multi-modal Artificial Intelligence*. *Advanced Science* (Co-first author).
 - Tianyi Xu, **Tianshuo Wei**, Ye Wei.(2026). *Feedback Is The Key for Automated Survey Generation*. *The 64th Annual Meeting of the Association for Computational Linguistics (ACL)* (Co-first author).
 - Fei Zhang, **Tianshuo Wei**, Ye Wei.(2026). *Bloom-Eval: A Hierarchical Evaluation Benchmark for Automatic Survey Generation Based on Bloom's Taxonomy*. *The 64th Annual Meeting of the Association for Computational Linguistics (ACL)* (Co-first author).
 - Maolin Wang, **Tianshuo Wei**, Xiangyu Zhao.(2025). *DANCE: Resource-Efficient Neural Architecture Search with Data-Aware and Continuous Adaptation*. *International Joint Conference on Artificial Intelligence (IJCAI)* (Co-first author) .
 - Binhao Wang, Maolin Wang, **Tianshuo Wei**, Xiangyu Zhao.(2025). *SPARK: Adaptive Low-Rank Knowledge Graph Modeling in Hybrid Geometric Spaces for Recommendation*. In *ACM International Conference on Information and Knowledge Management (CIKM)* (Co-first author).
 - Maolin Wang, Zhiqi Li, **Tianshuo Wei**, Xiangyu Zhao.(2026). *T-GINEE: A Tensor-based Multi-Graph Representation Learning*. *International Conference on Learning Representations (ICLR)*.
- Under Review
- Maolin Wang, Zhiqi Li, **Tianshuo Wei**, Xiangyu Zhao.(2026). *Unified Multimodal Recommendation via Tensor-level Regularization under Modality Incompleteness*(Submitted to ACM SIGIR).
 - Maolin Wang, Pengzhi Zhong, **Tianshuo Wei**, Xiangyu Zhao.(2026). *TrojanScope: Interpretable Backdoor Detection for Time Series Forecasting*(Submitted to IJCAI).
 - Maolin Wang, Xuhui Chen, **Tianshuo Wei**, Xiangyu Zhao.(2025). *RC-BiT: Rank-1 Compensated Binary Transformer*(Submitted to DAC).
 - Maolin Wang, Yanxin Chen, **Tianshuo Wei**, Xiangyu Zhao.(2025). *Mixture of Experts for Cross-Modal Feature Fusion in Sequential Recommendation Systems*(Submitted to KDD).
 - Maolin Wang, Binhao Wang, **Tianshuo Wei**, Xiangyu Zhao.(2025). *SONATA: Synergistic Coreset Informed Adaptive Temporal Tensor Factorization*(Submitted to ICML).
 - Maolin Wang, Yanxin Chen, **Tianshuo Wei**, Xiangyu Zhao.(2025). *Mixture of Experts for Cross-Modal Feature Fusion in Sequential Recommendation Systems*(Submitted to SIGKDD).

Work Experiences

- 2022.04–
2023.09
- Full-time Data Analyst, Lenovo (Shanghai) Electronic Technology Co., Ltd**
- Developed web crawlers using Python (BeautifulSoup, Scrapy) to extract and aggregate large volumes of customer data from multiple e-commerce platforms
 - Implemented machine learning models (Random Forest, XGBoost) for analyzing sales patterns
 - Applied NLP techniques to mine user feedback data, identifying key satisfaction factors through sentiment analysis and topic modeling
 - Created automated data visualization dashboards that transformed complex analysis into actionable business insights
- 2020.09–
2022.04
- Full-time Automation Development, Lenovo (Shanghai) Electronic Technology Co., Ltd**
- Developed automation projects using Python, implemented test scenarios automation with Selenium and Pytest frameworks, reducing test cycle by 70%
 - Conducted data mining and visualization analysis on test results to identify performance bottlenecks and anomaly patterns
 - Proposed system optimization solutions based on analysis results, improving system performance by 35% and stability by 40%

Extracurricular Activities

- 2018.09–
2020.06
- Director, PR Department, Nottingham CSSA (Chinese Students and Scholars Association)**
- Led a team of 30 members to orchestrate all facets of the Spring Festival events, including marketing, logistics, and on-site management, successfully engaging over 3,000 local people per event.
 - Served as a campus ambassador to introduce campus culture and the surrounding facilities to freshmen through live streaming resulting in a 70% reduction in common queries from new students
- 2017.09–
2018.06
- Member, Pixel Department, Library, University of Nottingham Ningbo China**
- Co-developed a WeChat mini-program for book searches, independently responsible for the front-end UI/UX design and implementation using WXML and WXSS.
 - Directed, filmed, and edited a lab safety promotional video using Adobe Premiere Pro. The video was adopted as official training material for all first-year science students.

Skills & Tools

Programming Python, MATLAB, R
Algorithms Linear/Logistic Regression, Decision Trees, K-Means Clustering, PCA
Tools Advanced Excel, SQL

Honors & Awards

Cityu MSDS Outstanding Dissertation Award (1/296)
IJCAI Volunteer for IJCAI 2025 Student Volunteer Program
UNNC Nottingham Advantage Award — Graduation Promotion and Memento Project
UNNC Nottingham Advantage Award — Exchange Season